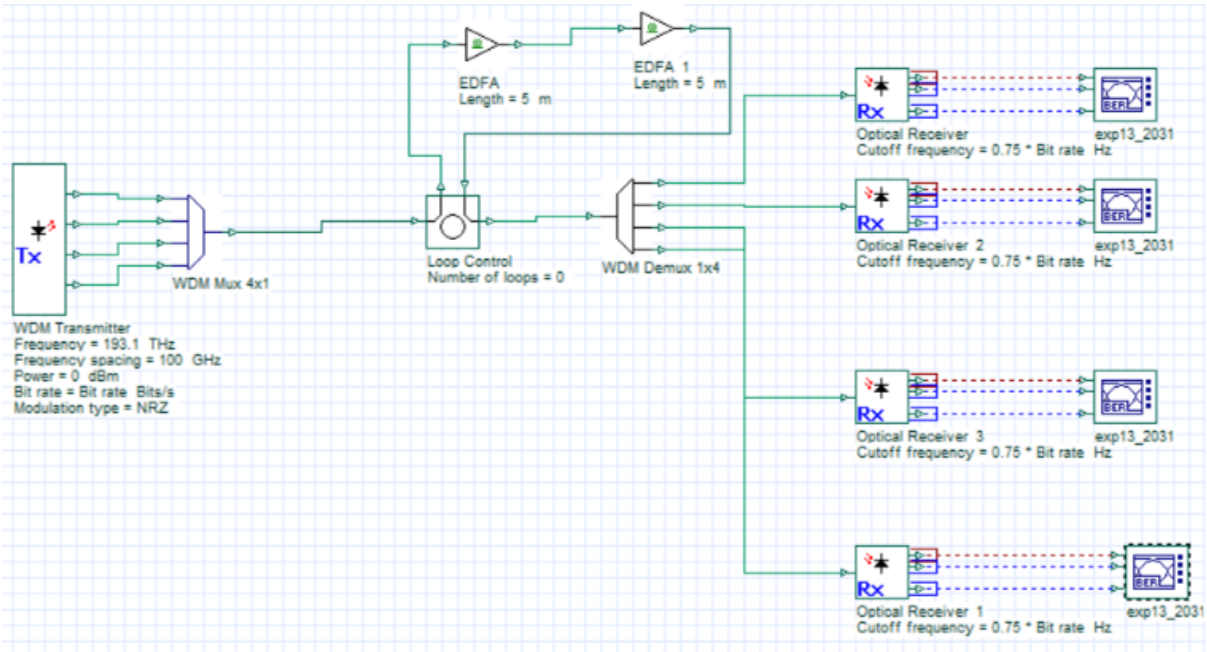
EXPERIMENT 13 : PERFORMANCE EVALUATION OF DENSE WDM MULTIPLEXING AND DEMULTIPLEXING USING OPTISYSTEM

OBJECTIVES :

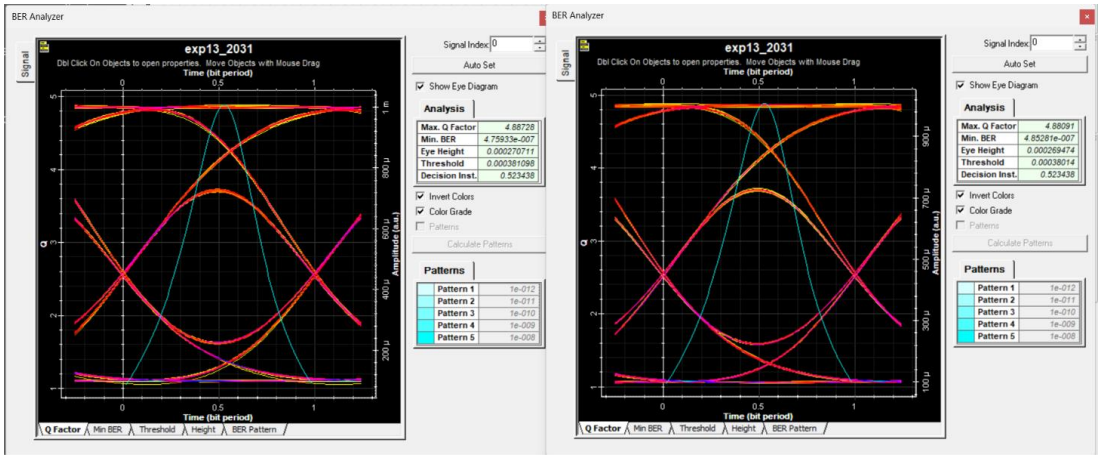
1. To design and simulate a Dense Wavelength Division Multiplexing (DWDM) system in OptiSystem and analyze its performance under different test conditions.
2. To analyze the effect of wavelength (channel) spacing, fiber length (loop count) and channel power on BER, Q-factor, OSNR and crosstalk.
3. To compare modulation formats (NRZ, RZ, QPSK) and determine the optimal optical amplifier (EDFA) placement for long-haul DWDM transmission.

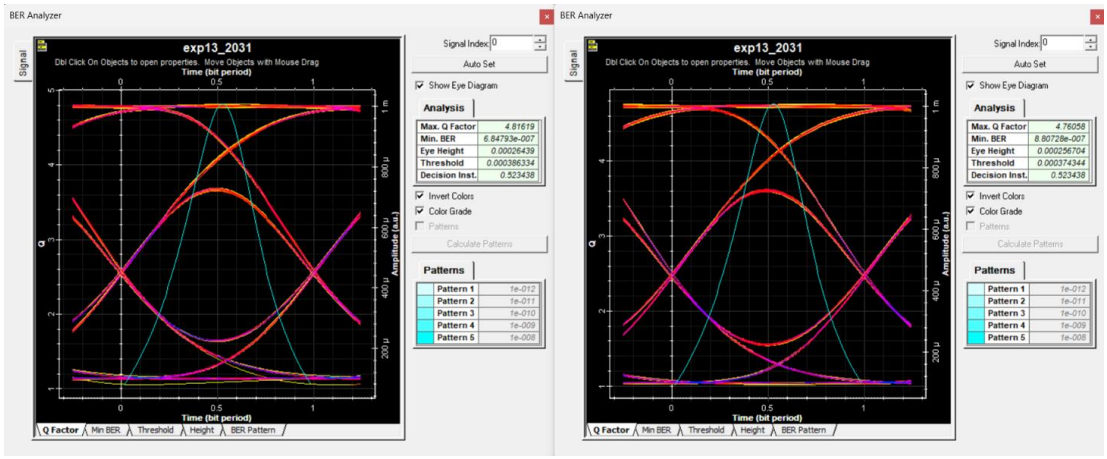
SOFTWARE USED : OPTISYSTEM SOFTWARE

CIRCUIT DIAGRAM:



RESULTS:





REFLECTION:

I learned to design and simulate a DWDM system using Opti system and evaluate the effects of channel spacing, fiber length, modulation formats, and amplifier placement on system performance.

I understood the trade-off between spectral efficiency and transmission quality by analysing BER, Q factor ,OSNR, and crosstalk.

RUBRICS:

Objective		Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
System Design & DWDM Setup		Correct selection and connection of OptiSystem components for the DWDM link (CW Laser Array, Modulator/Pulse Generator, WDM MUX, Optical Fiber with Loop Control, EDFA/Raman Amplifier, WDM DEMUX, PIN/APD Photodetectors, BER Analyzer, Optical Spectrum Analyzer, Optical Power Meter).	All transmiVer, MUX, fiber loop, amplifier, DEMUX and receiver blocks are correctly selected, connected in sequence and configured without errors.	All components are present with minor connection or layout errors, but the DWDM link remains functional.	Component selection is partially complete; missing analyzers or amplifier/loop errors impede signal flow.	Incorrect component selection or improper setup leading to complete simulation failure.	20%
Parameter Configuration & Test Case Execution		Accurate setup of ITUgrid frequencies (193.1 THz onwards), channel spacing (100/50/25 GHz), loop count (1/5/10 loops of 50 km), modulation (NRZ/RZ/QPSK), EDFA spacing (50/100/150 km), channel power (0–10 dBm), fiber aVenuation (0.2 dB/km), dispersion (16 ps/nm/km) and amplifier noise figure (5 dB).	All parameters are accurately set and all five test cases are executed completely.	Parameters are mostly accurate with minor seVing discrepancies; one test case is incomplete.	Basic parameters are configured, but multiple test cases or parameter values are missing.	Incorrect parameter seVings or failure to run the required test cases.	20%
Measurement & Performance Monitoring		Accurate extraction and recording of BER, Q-factor, OSNR, crosstalk and power penalty for every test case, using the BER Analyzer, Optical Spectrum Analyzer and Optical Power Meter.	Precise data extraction across all test cases, neatly tabulated with complete BER, Qfactor and OSNR values and clear spectra and eye diagrams.	Data is collected with minor omissions in BER, Q-factor, OSNR or crosstalk readings.	Partial data collection with incomplete tables or missing spectrum/eye diagram captures.	Inaccurate, incomplete or missing data and measurements.	20%
Performance Analysis		Detailed evaluation of how spacing, distance, modulation format, amplifier placement and channel power affect crosstalk, dispersion, ASE	Comprehensive analysis correctly relating each parameter to BER, Qfactor and OSNR with sound physical explanations and trade-offs.	Clear analysis identifying basic trends in BER, Qfactor and OSNR for most test cases.	Basic aVempt at result analysis without explaining physical mechanisms such as crosstalk, dispersion or	Erroneous or missing analysis of simulation results.	20%

		noise, SPM/XPM nonlinearity and spectral efficiency.			nonlinear effects.		
Report & Conclusion		Identification of optimal spacing, amplifier interval, modulation format and power level; structure and clarity of the report.	Report features clear spectra and eye diagrams, complete result tables and insightful conclusions identifying optimal DWDM design parameters.	Satisfactory report with relevant displays and a sound concluding summary.	Incomplete report with missing graphs, readings or weak conclusions.	Disorganized or incomplete report lacking required diagrams and valid conclusions.	20%

CONCLUSION:

Using OptiSystem to simulate DWDM systems enables detailed performance evaluation under various conditions. The insights from these test cases help optimize DWDM designs for high-capacity, long-haul communication networks.